Chapter 5: Model Selection

5.1 Bias and variance

A model with high bias is too simple to capture the pattern in the data. It underfits: both its training error and its validation error are high. A model with high variance is too flexible. It overfits: its training error is very low but its validation error is high, because it has memorized noise in the training set.

As model complexity increases, the training error keeps decreasing. The validation error first decreases and then increases again. The best model is the one near the minimum of the validation error curve. Figure 5.1 on the next page shows this pattern for polynomial regression.

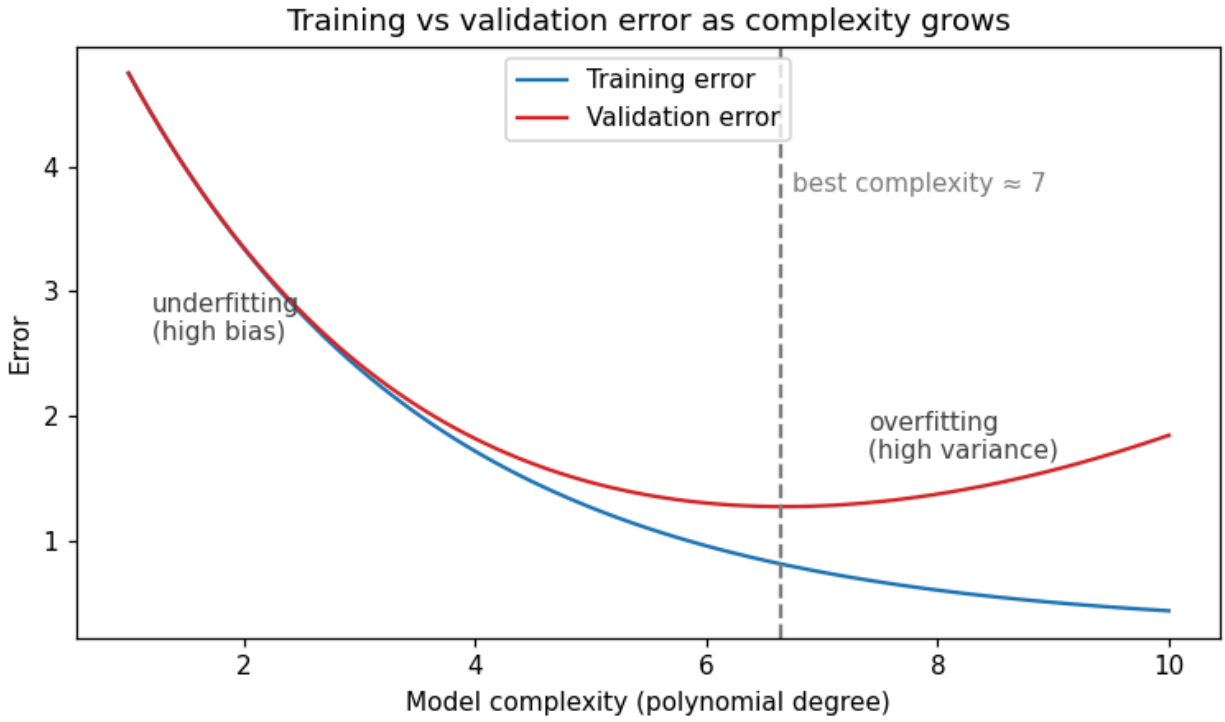


Figure 5.1: Training and validation error as model complexity grows. Training error falls steadily; validation error is U-shaped.

The gap between training and validation error is a direct measure of overfitting. When the gap is wide, reduce model complexity, add regularization, or collect more data. Never choose the complexity using the test set.

5.2 Cross-validation

When data is limited, a single validation split gives a noisy estimate. k-fold cross-validation splits the data into k equal folds. Each fold is used once as the validation set while the other k-1 folds train the model. The k validation scores are averaged to estimate performance. Typical choices are k = 5 or k = 10; leave-one-out is the extreme case k = n.

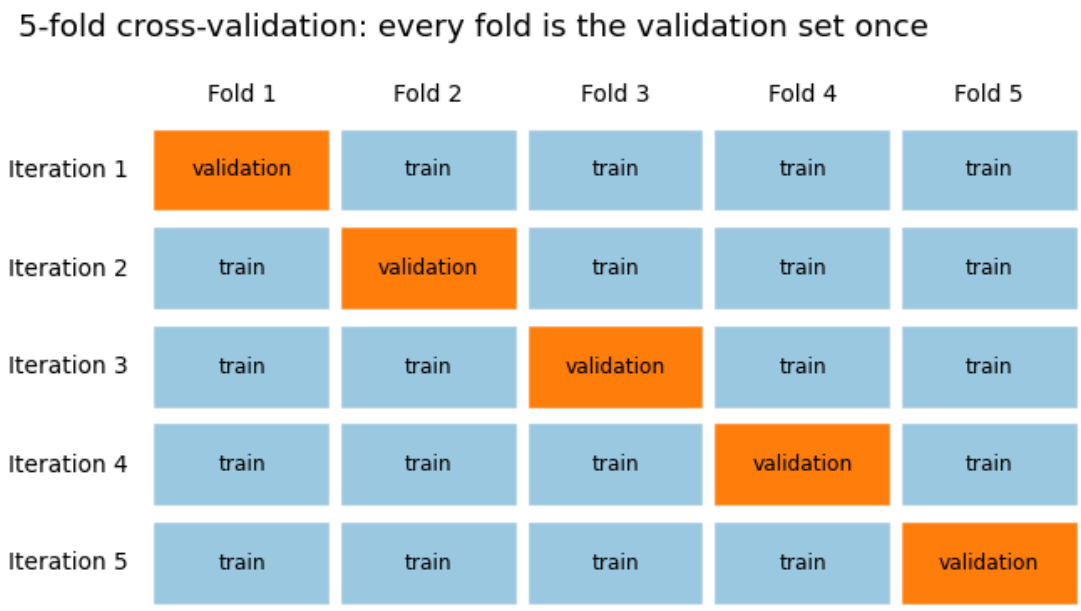


Figure 5.2: 5-fold cross-validation. In each iteration a different fold is held out for validation.